

X1
DigitalCarrierJack.SchDoc

Encoder1
Encoder1.SchDoc

Encoder2
Encoder2.SchDoc

Encoder3
Encoder3.SchDoc

Connector
Connectors.SchDoc

DIG_IO
PowerSupply
DIG_Res
Status
I2C
CPLD_IO

DIG_IO
PowerSupply
DIG_Res
Status
I2C
CPLD_IO

Encoder_Signals_1
Encoder_Signals_2
Encoder_Signals_3

Encoder_Signals

Encoder_Signals

Encoder_Signals

GND
GND



Revision

Serial
Serial

UltraZohm_DigitalVoltage
Project

Designer

Date

Title		TopSheet.SchDoc		Technische Hochschule Nürnberg Georg Simon Ohm Fürther Straße 250 90429 Nürnberg GERMANY		
Size:	A4	Revision:	1v00			
Date:	15.02.2022	Time:	15:15:54	Sheet 1 of 6	Author: Michaela Hlatky	
Project: UltraZohm_Digital_Incremental_Encoder_v2.PrjPCB						

A

B

C

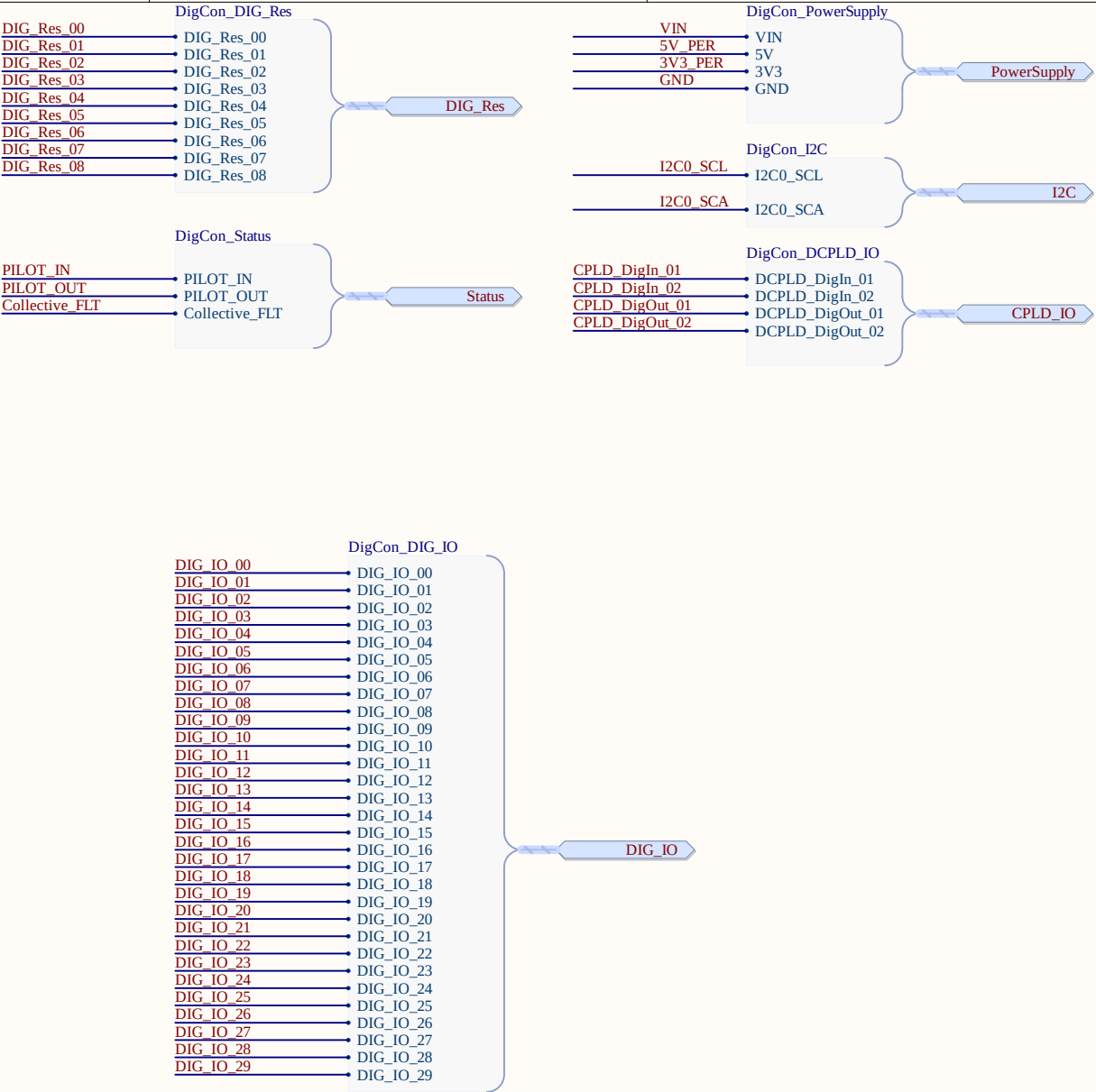
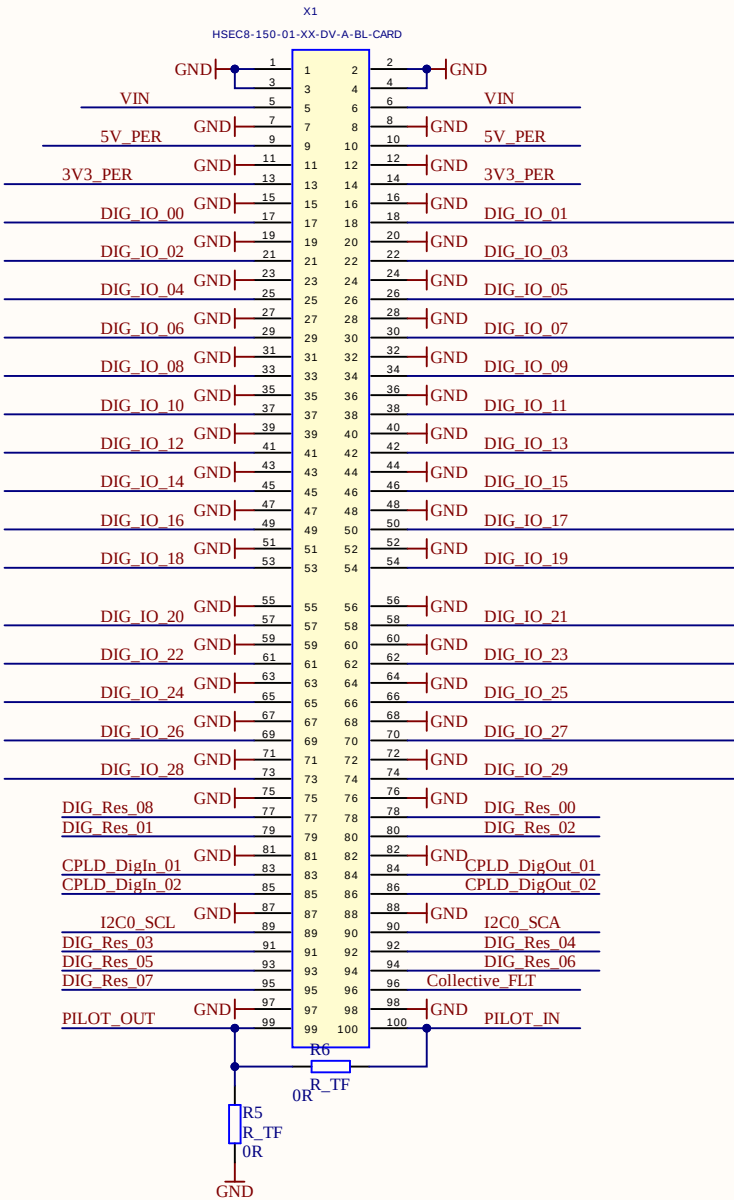
D

A

B

C

D



A

B

C

D

A

B

C

D

DigCon_DIG_IO

DIG_IO_00 → DIG_IO_00
DIG_IO_01 → DIG_IO_01
DIG_IO_02 → DIG_IO_02
DIG_IO_03 → DIG_IO_03
DIG_IO_04 → DIG_IO_04
DIG_IO_05 → DIG_IO_05
DIG_IO_06 → DIG_IO_06
DIG_IO_07 → DIG_IO_07
DIG_IO_08 → DIG_IO_08
DIG_IO_09 → DIG_IO_09
DIG_IO_10 → DIG_IO_10
DIG_IO_11 → DIG_IO_11
DIG_IO_12 → DIG_IO_12
DIG_IO_13 → DIG_IO_13
DIG_IO_14 → DIG_IO_14
DIG_IO_15 → DIG_IO_15
DIG_IO_16 → DIG_IO_16
DIG_IO_17 → DIG_IO_17
DIG_IO_18 → DIG_IO_18
DIG_IO_19 → DIG_IO_19
DIG_IO_20 → DIG_IO_20
DIG_IO_21 → DIG_IO_21
DIG_IO_22 → DIG_IO_22
DIG_IO_23 → DIG_IO_23
DIG_IO_24 → DIG_IO_24
DIG_IO_25 → DIG_IO_25
DIG_IO_26 → DIG_IO_26
DIG_IO_27 → DIG_IO_27
DIG_IO_28 → DIG_IO_28
DIG_IO_29 → DIG_IO_29

DIG_IO

Encoder_Signals_CH3

A_encoder → A_encoder3
B_encoder → B_encoder3
Z_encoder → Z_encoder3

Encoder_Signals_3

Encoder_Signals_CH2

A_encoder → A_encoder2
B_encoder → B_encoder2
Z_encoder → Z_encoder2

Encoder_Signals_2

Encoder_Signals_CH1

A_encoder → A_encoder1
B_encoder → B_encoder1
Z_encoder → Z_encoder1

Encoder_Signals_1

DIG_Res

DigCon_DIG_Res

DIG_Res_00 → DIG_Res_00
DIG_Res_01 → DIG_Res_01
DIG_Res_02 → DIG_Res_02
DIG_Res_03 → DIG_Res_03
DIG_Res_04 → DIG_Res_04
DIG_Res_05 → DIG_Res_05
DIG_Res_06 → DIG_Res_06
DIG_Res_07 → DIG_Res_07
DIG_Res_08 → DIG_Res_08

CPLD_IO

DigCon_DCPLD_IO

DCPLD_DigIn_01 → CPLD_DigIn_01
DCPLD_DigIn_02 → CPLD_DigIn_02
DCPLD_DigOut_01 → CPLD_DigOut_01
DCPLD_DigOut_02 → CPLD_DigOut_02

Status

DigCon_Status

PILOT_IN → PILOT_IN
PILOT_OUT → PILOT_OUT
Collective_FLT → Collective_FLT

DIG_IO_00 01
DIG_IO_01 04
DIG_IO_03 07
DIG_IO_05 10
DIG_IO_07 13
DIG_IO_09 16
DIG_IO_11 19
DIG_IO_13 22
DIG_IO_15 25
DIG_IO_17 28
01 02
04 05
07 08
10 11
13 14
16 17
19 20
22 23
25 26
28 29

GND

X2

01 02
04 05
07 08
10 11
13 14
16 17
19 20
22 23
25 26
28 29

Pin Header

X4

VIN_fused 01
GND 03
5V_PER_fused 05
GND 07
3V3_PER_fused 09
GND 11
01 02
03 04
05 06
07 08
09 10
11 12
02 VIN_fused
04 GND
06 5V_PER_fused
08 GND
10 3V3_PER_fused
12 GND

Pin Header

DIG_IO_21 01
DIG_IO_23 04
DIG_IO_25 07
DIG_IO_27 10
DIG_IO_29 13
DIG_Res_07 16
CPLD_DigIn_01 19
CPLD_DigIn_02 22
I2C0_SCL 25
PILOT_OUT 28
01 02
04 05
07 08
10 11
13 14
16 17
19 20
22 23
25 26
28 29
02 DIG_IO_22
05 DIG_IO_24
08 DIG_IO_26
11 DIG_IO_28
14 DIG_Res_08
17 DIG_Res_08
20 CPLD_DigOut_01
23 CPLD_DigOut_02
26 I2C0_SCA
29 PILOT_IN

GND

X3

01 02
04 05
07 08
10 11
13 14
16 17
19 20
22 23
25 26
28 29

Pin Header

F1

VIN → VIN_fused
Littelfuse_PTC_1812
F2
5V_PER → 5V_PER_fused
Littelfuse_PTC_1812
F3
3V3_PER → 3V3_PER_fused
Littelfuse_PTC_1812

DigCon_I2C

I2C0_SCL → I2C0_SCL
I2C0_SCA → I2C0_SCA

I2C

DigCon_PowerSupply

VIN → VIN
5V → 5V_PER
3V3 → 3V3_PER
GND → GND

PowerSupply

Title Connectors.SchDoc

Size: A4

Revision: 1v00

Date: 15.02.2022 Time: 15:15:54

Sheet 3 of 6

Author: Michaela Hlatky

Project: UltraZohm_Digital_Incremental_Encoder_v2.PrjPCB

Technische Hochschule Nürnberg Georg Simon Ohm
Fürther Straße 250
90429 Nürnberg
GERMANY

TECHNISCHE HOCHSCHULE NÜRNBERG
GEORG SIMON OHM

A

B

C

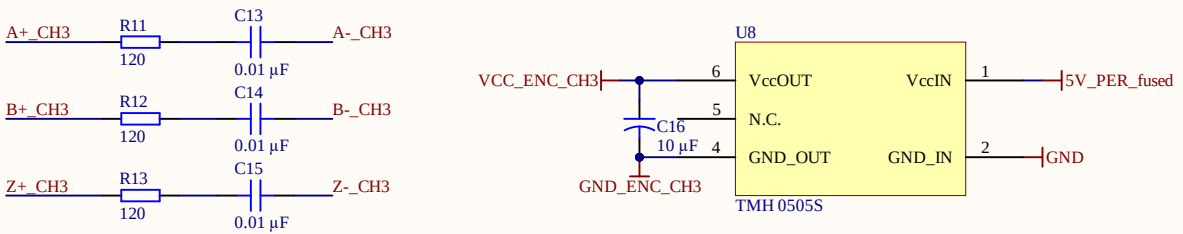
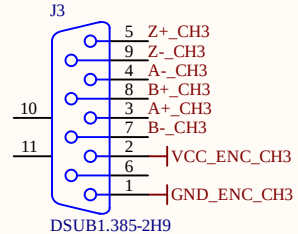
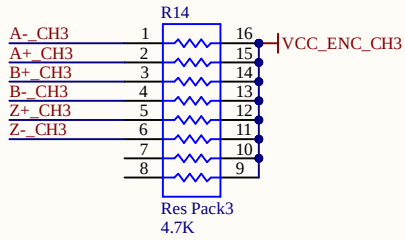
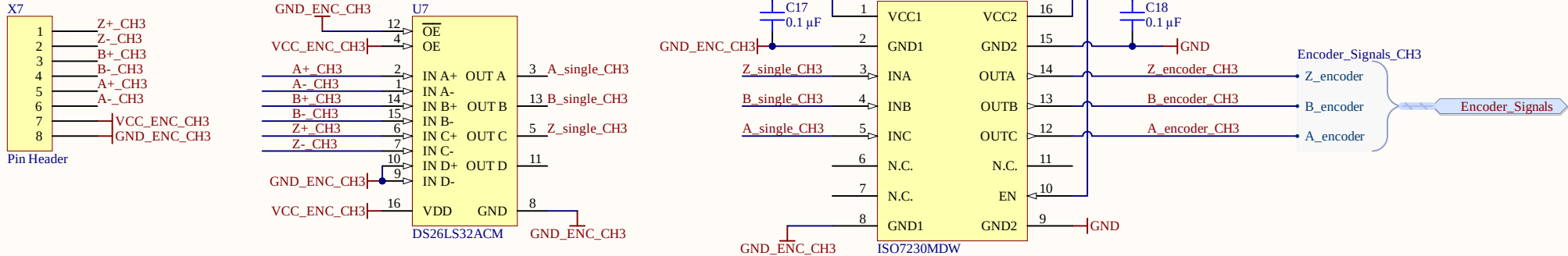
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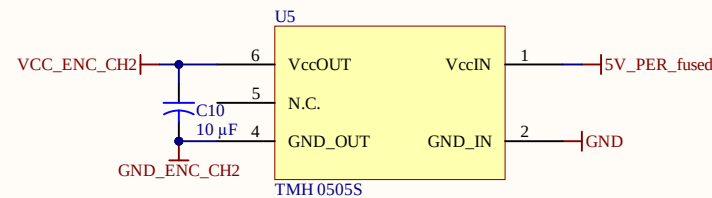
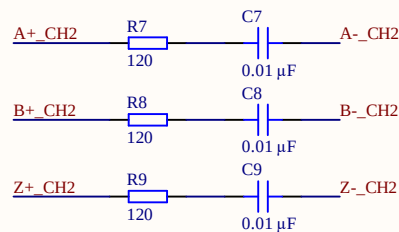
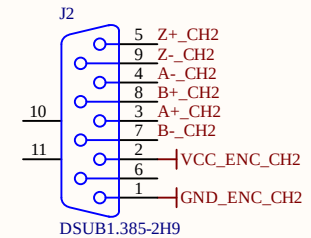
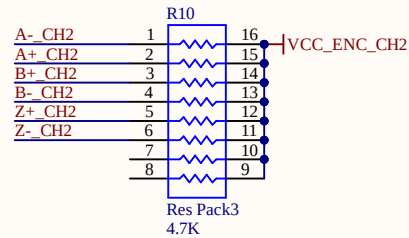
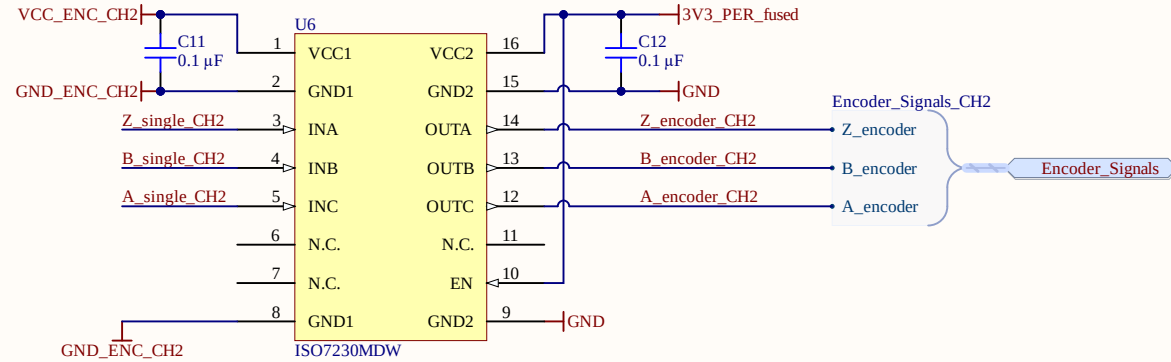
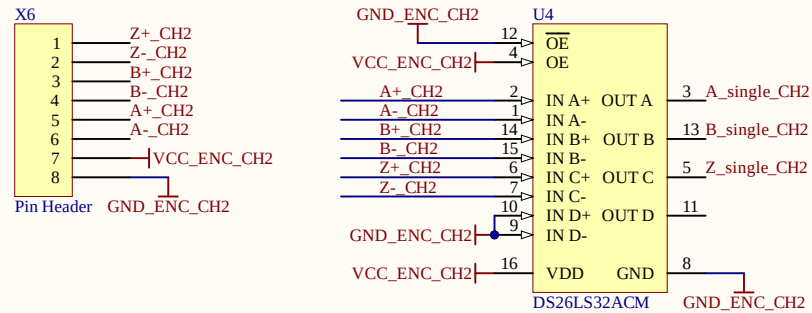
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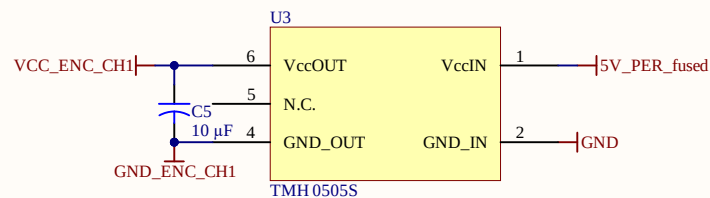
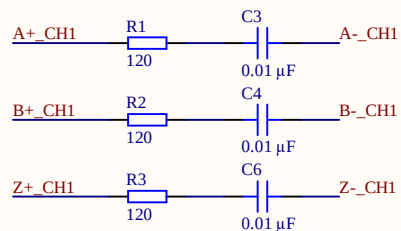
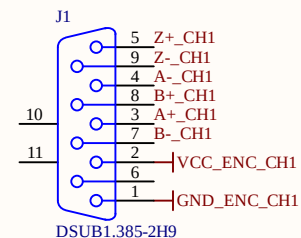
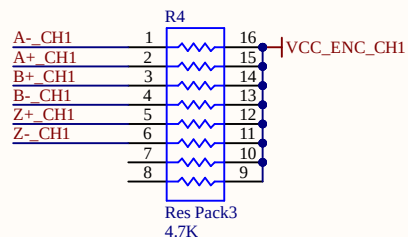
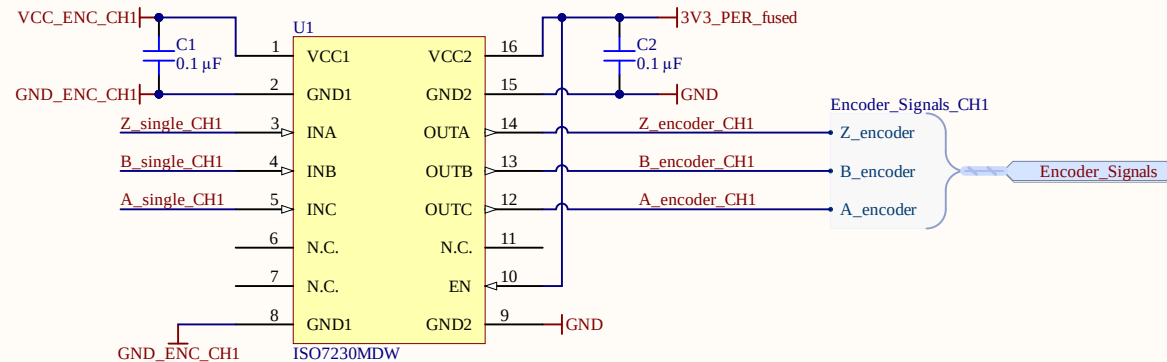
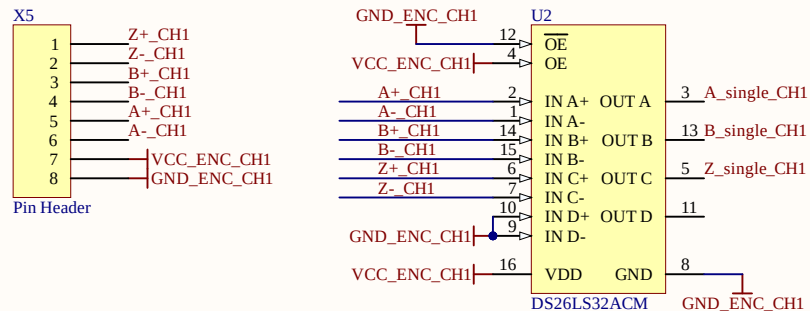
D



Label	Pin	Encoder	Color(Kübler)
U0+	Pin3	0+	blue
U0-	Pin4	0-	red
U1+	Pin8	A+	green
U1-	Pin7	A-	yellow
U2+	Pin5	B+	grey
U2-	Pin9	B-	pink
VCC_ENC	Pin2	UB	brown
GND_ENC	Pin1	0V	white
NC	Pin6	--	



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GND_ENC	Pin1	0V	white
NC	Pin6	--	